

Annotated Bibliography: Dimensional Emergence and Information

1. Holographic Principle ('t Hooft, Susskind)

- **'t Hooft (1993)** – *“Dimensional reduction in quantum gravity”* ¹. Introduces the idea that combining quantum mechanics with gravity implies the true fundamental description of the universe need not be 3+1 dimensional. He shows that observable degrees of freedom in a gravitational setting can be described as Boolean variables on a two-dimensional lattice evolving in time ¹. In other words, spatial volume information is encoded on a surface, suggesting that an extra dimension “emerges” to accommodate information and prevent collapse. This supports the notion that spatial dimensions might arise as information-stabilizing structures. Open questions remain about how this principle is derived from first principles and how it applies beyond black-hole horizons (e.g. in cosmological settings) ².
- **Susskind (1995)** – *“The World as a Hologram”* ³. Frames the universe as a holographic image: a 3D region is fully encoded by information on its 2D boundary. He argues that there is roughly one fundamental bit per Planck-area on the boundary that suffices to describe the entire 3D volume ³. This reinforces the holographic picture that higher dimensions “emerge” from lower-dimensional information content. The implication for dimensional emergence is that adding a spatial dimension can be understood as unfolding a lower-dimensional information manifold. It remains unresolved how this principle manifests in de Sitter (expanding) cosmology or in dynamical spacetimes, as most examples (AdS/CFT, black holes) assume special settings.
- **Bousso (2002)** – *“The holographic principle”* ². Reviews evidence that the maximum information content of any spacetime region scales with its boundary area ($\sim 10^{69}$ bits per m^2) rather than its volume ². Bousso emphasizes that this “area-information” relation is a universal geometric constraint and asserts it must originate in the fundamental quantum-gravitational degrees of freedom ². For dimensional emergence, this suggests that spatial surfaces play a stabilizing role in storing information. Crucially, he notes that *“It has yet to be explained”* how this relation arises from microphysics ², highlighting an open question: deriving the holographic bound from a concrete theory. Another open issue is how holography extends to realistic cosmology (e.g. positive cosmological constant).

2. Integrated Information Theory (Tononi)

- **Tononi (2004)** – *“An information integration theory of consciousness”* ⁴. Proposes that consciousness corresponds to a system’s capacity to integrate information. He defines a measure Φ (“phi”) quantifying the irreducible information generated by a complex of elements, above and beyond that of its parts ⁴. In IIT, a high-dimensional causal structure emerges (a “maximally irreducible conceptual structure”) to support large Φ , meaning the system behaves as a unified whole. In the context of dimensional emergence, one could view each spatial or temporal dimension as enabling extra degrees of freedom for information integration. For example, higher-dimensional connectivity may be needed to sustain high Φ . Open issues include how to compute Φ in realistic networks and whether maximizing Φ uniquely selects certain physical substrates. Critics also note IIT’s computational intractability: calculating Φ scales super-exponentially and is only feasible for tiny systems ⁵.

- **Oizumi et al. (2014)** – “From the phenomenology to the mechanisms of consciousness: Integrated Information Theory 3.0” ⁶ ⁷ . This updated IIT formulation starts from axioms of conscious experience and defines **intrinsic information** (“differences that make a difference” internal to a system) and **integrated information** as information specified by the whole that cannot be reduced to independent parts ⁶ . They identify a system’s experience as a *maximally irreducible conceptual structure* (MICS), with Φ^{Max} measuring its size ⁷ . The relevance to dimensional emergence is conceptual: IIT suggests that higher-order integrated structures (analogous to emergent dimensions) exist when parts combine in an irreducible way. The paper also notes limitations: the formalism assumes small, discrete networks and faces a combinatorial explosion for larger systems ⁵ . In particular, finding the “grain” (optimal spatial/temporal scale) where Φ is maximized is nontrivial; in general “the macro may emerge over the micro, just like the whole may emerge above the parts” ⁸ . This hints that coarse-graining (adding effective dimensions or layers) could be informationally optimal.

3. Entropic Gravity (Verlinde)

- **Verlinde (2011)** – “On the origin of gravity and the laws of Newton” ⁹ . Derives Newton’s laws from information-theoretic and thermodynamic principles: space is considered emergent and gravity is an **entropic force** resulting from changes in information when matter moves relative to a holographic screen. He writes: “Gravity is explained as an entropic force caused by changes in the information associated with the positions of material bodies” ⁹ . This posits that what we perceive as a gravitational “dimension” is really a gradient of entropy/information. In terms of dimensional emergence, gravity itself arises because the system tends to arrange into higher dimensions to maximize entropy, stabilizing information distribution. Extensions by Verlinde (2016) argue that spacetime and gravity emerge from the entanglement structure of a microscopic theory, and predict an extra “dark gravity” force from entropy displacement ¹⁰ ¹¹ . Open questions include experimental tests and critiques: for example, Kobakhidze (2010) argues that precision neutron experiments are incompatible with a purely entropic origin of gravity ¹² . Verlinde’s ideas also face debate about whether all gravitational phenomena (e.g. in de Sitter cosmology or galactic dynamics) can be accounted for.
- **Kobakhidze (2011)** – “Gravity is not an entropic force” ¹² . Presents a counter-argument by showing that ultra-cold neutron experiments in Earth’s gravity (neutron bound states) seem inconsistent with Verlinde’s entropic-gravity predictions. In particular, Kobakhidze concludes that observed quantum states of neutrons conflict with an entropic interpretation of gravity ¹² . This highlights an open controversy: Verlinde’s model remains speculative and faces challenges from laboratory tests. Further research is needed to reconcile entropic-gravity ideas with precision data or to delimit their domain of validity.

4. Causal Dynamical Triangulations (CDT)

- **Ambjørn, Jurkiewicz, & Loll (2010)** – “Causal dynamical triangulations and the quest for quantum gravity” ¹³ . Reviews the CDT approach, which constructs spacetime by summing over simplicial geometries with a fixed causal structure. They emphasize that CDT, with minimal assumptions, “has demonstrated that a classical universe can be generated dynamically from Planckian quantum fluctuations” ¹³ . Indeed, large-scale behavior of the CDT ensemble yields a de Sitter-like four-dimensional spacetime, even though the microscopic building blocks are discrete and quantum. This shows spatial and temporal dimensions emerging as collective, stable phenomena. Relevance to dimensional emergence is direct: CDT explicitly demonstrates how extended dimensions can emerge from lower-level information (the connectivity of simplices). Open issues

they note include the need to tune parameters to approach the continuum limit and extract Newton's law from the theory ¹⁴ ¹⁵ .

- **Ambjørn, Jurkiewicz, & Loll (2005)** – “*Spectral Dimension of the Universe*” ¹⁶ . Presents a key result: the effective (spectral) dimension of the CDT spacetime depends on scale. Their simulations show that **at large scales the universe is four-dimensional, but at very short (Planckian) scales it appears two-dimensional** ¹⁶ . This “dynamical dimensional reduction” suggests that near the Planck scale information behaves as if living in lower dimensions (which may regularize quantum gravity). In the context of information collapse, this implies that adding dimensions only occurs at scales where needed to support stable information flow; at very small scales the system effectively has fewer degrees of freedom. An open question is whether this reduction is universal to all quantum gravity approaches (it also appears in Horava–Lifshitz models) and how it connects to the holographic behavior of black holes.
- **Ambjørn et al. (2010, Outlook)** – “*Open issues and outlook*” ¹⁷ . They report that at short distances the emergent quantum spacetime is “nothing like a four-dimensional classical manifold” ¹⁷ . For example, the average curvature (Regge curvature) diverges at the lattice level, and spacetime has an anomalous (fractal) spectral dimension ¹⁷ . These findings illustrate that while large-scale 4D geometry emerges, the microscopic structure is highly non-classical. For dimensional emergence theories, this raises open questions: Which features of spacetime must be built in (fixed topology, causality, etc.) and which truly emerge? Also, identifying a continuum UV fixed point (if any) remains a key challenge.

5. Heimburg–Jackson Soliton Theory (nerve impulses)

- **Heimburg & Jackson (2005)** – “*On soliton propagation in biomembranes and nerves*” ¹⁸ ¹⁹ . Propose that action potentials are not purely electrical (ion diffusion) but are *mechanical density pulses* (solitons) propagating in the nerve membrane. They show that lipid membranes near their chain-melting transition can support localized, distortionless compression waves ¹⁸ . Experimentally, nerve pulses indeed exhibit reversible thermodynamics: heat released then reabsorbed during a spike. The soliton model naturally accounts for this (an adiabatic, reversible pulse) ²⁰ . From an information perspective, this suggests that biological “dimensions” (membrane states and propagating modes) emerge to stabilize neural information transmission in a minimally-dissipative way. The model has testable predictions: e.g. anesthetics depress the membrane melting point (affecting pulse propagation) and hydrostatic pressure raises it, so applying pressure reverses anesthesia ²¹ . This points to an intimate link between membrane phase behavior and information-bearing nerve signals. Open issues include how soliton pulses deal with neuron branching and synapses, and why Hodgkin–Huxley (HH) ion-channel theory still explains most electrophysiology. The soliton theory remains controversial, and integration with known ion-channel dynamics is an area of active investigation.
- **Heimburg & Jackson (2007)** – “*The thermodynamics of general anesthesia*” ²² . Extend the membrane-soliton framework to anesthesia: they treat anesthetics as freezing-point depressants of the lipid membrane. Their analysis predicts that **pressure increases and lowered pH reverse anesthesia**, in agreement with clinical findings ²² . The basic thermodynamic relation they derive shows the anesthetic dose (ED50) is directly proportional to how much the agent lowers the membrane's phase transition temperature. This explains the known Meyer–Overton correlation and why high pressure can counteract anesthetics. In terms of dimensions, this implies that altering an external “pressure dimension” (either literal pressure or chemical potential) changes the internal informational state space of the membrane. The success of this

model in explaining reversible anesthesia supports the idea that the physical phase of membranes is a carrier of neural information. Remaining questions include quantitative comparisons with diverse anesthetics and reconciling this membrane view with molecular targets (ion channel binding).

6. Acoustic Metamaterials

- **Park et al. (2021)** – “*Ultraslow medium with an acoustic membrane-like undamped dynamic vibration absorber...*”²³ ²⁴. Demonstrates an engineered metamaterial in which a thin membrane array with coiled resonators achieves **an effective density ~2000× that of air**, producing an ultraslow sound medium (sound speed below 7.5 m/s)²³. The design relies on coupled resonances (Fano interference) that give rise to extreme dynamic mass loading: at a certain frequency the membrane displacement is canceled by the rings, yielding nearly infinite inertia²⁴. Such systems can exhibit negative effective mass density and even imaginary wave vectors around resonance²⁴. The relevance to dimensional emergence is analogy: the metamaterial's effective acoustic dimension (wave speed and impedance) emerges from subwavelength structural resonances, illustrating how added “internal structure” can stabilize or slow wave information. This example shows how new degrees of freedom (the resonant elements) create an emergent acoustic environment with novel information-carrying capacity. Open questions include bandwidth (these effects are narrowband around resonances) and loss (viscous damping) which limit practical applications.
- **Ma & Sheng (2016)** – “*Acoustic metamaterials: from local resonances to broad horizons*”²⁵. This review surveys developments in metamaterials with unusual effective parameters. It notes that acoustic metamaterials permit **effective mass density (ρ) and bulk modulus (κ) to take negative, near-zero, or divergent values** in effective-medium theory²⁵. Such exotic parameters enable behaviors like negative refraction, superlensing, and cloaking for sound²⁶. In information terms, these structures create additional “synthetic” dimensions in wave propagation: for example, a negative density medium can invert the phase response of waves, effectively giving an extra sign degree of freedom. The review also highlights resonance effects (e.g. membranes, Helmholtz resonators) as the mechanism behind these parameters. A key open issue is scalability: metamaterial effects often appear only near narrow frequency bands, so extending them to broadband, low-loss regimes is an ongoing research goal.

7. Hyperbaric Neurophysiology (HPNS, Pressure Effects)

- **Jain (1994)** – “*High-pressure neurological syndrome (HPNS)*”²⁷. Summarizes clinical observations that divers descending below ~150 m experience tremors, myoclonus, vertigo, cognitive impairment and EEG disturbances²⁷. HPNS is attributed to high pressure increasing neuronal excitability via membrane compression and altered ion-channel function²⁸. In information terms, extreme pressure perturbs the “membrane dimension” of neurons, disrupting stable signaling. Notably, certain anesthetic gases alleviate HPNS symptoms by reversing pressure-induced membrane changes (“pressure reversal” of narcosis)²⁹. This reversible anesthesia effect implies that the same physical substrate (membrane state) governs both narcosis and pressure syndromes. Key questions include the precise molecular mechanisms (e.g. NMDA vs. GABA systems) that underlie HPNS and its recovery, and how subjective perception (e.g. cognitive “time dilation” or slowing of information processing under pressure) relates to these physiological changes. The paradoxical pressure-reversal phenomenon in particular suggests deep links between environmental dimensions (pressure) and neural information dynamics²⁹.

Sources: The above entries draw on primary literature in holography ('t Hooft 1993; Susskind 1995; Bousso 2002), consciousness theory (Tononi 2004; Oizumi et al. 2014), entropic gravity (Verlinde 2011; Kobakhidze 2011; Verlinde 2017), causal triangulations (Ambjørn *et al.*, 2005; 2010), biophysical nerve models (Heimburg & Jackson 2005, 2007), acoustic metamaterials (Ma & Sheng 2016; Park *et al.* 2021), and hyperbaric physiology (Jain 1994). Citations mark key findings and emphasize how each framework suggests dimensions or structures emerge to stabilize information; they also note open issues and debates in each area.

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2 [hep-th/0203101] The holographic principle

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4 (PDF) An Information Integration Theory of Consciousness

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